

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) What is the mean of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 10
- 2) What is the median of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 9
- 3) The six observations 4, 6, 6, 8, 10, 14 have mean 8 and squared-deviation sum 64. What is their population variance? (1)
(A) $64/6$
(B) $64/5$
(C) $\sqrt{64/6}$
(D) 8
- 4) A distribution has mean 8, median 7 and positive standard deviation. What is the sign of Pearson's second skewness coefficient? (1)
(A) Zero
(B) Negative
(C) Undefined
(D) Positive
- 5) Which measure of central tendency is generally less affected by an extreme observation? (1)
(A) Arithmetic mean
(B) Median
(C) Midrange
(D) All equally affected

Section B: Theory questions (25 marks)

- 6) Find the five-number summary and interquartile range of 3, 5, 7, 8, 9, 11, 13, 16, 30. Exclude the median when forming the lower and upper halves. Use the 1.5 *IQR* rule to identify outliers and sketch a box plot. (3)
- 7) A table contains customer ID, age, monthly income and annual income. Give one suitable treatment for missing ages and explain how you would detect inconsistency between the two income fields. (2)
- 8) A filling machine is claimed to dispense a mean of 500 ml. A random sample of 36 bottles has mean 496 ml. The population standard deviation is known to be 12 ml. Perform a two-sided *z*-test at the 5% level. State the hypotheses, statistic, decision and conclusion. Use critical values ± 1.96 . (5)
- 9) Two judges rank six projects. Judge A assigns ranks (1, 2, 3, 4, 5, 6) and judge B assigns (2, 1, 4, 3, 6, 5). Calculate Spearman's rank correlation coefficient and interpret the result. (3)
- 10) Partition 10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 40 into three equal-frequency bins. Replace each observation by its bin mean and report the three resulting bin means. (2)
- 11) Test independence between training and passing an assessment using Pearson's chi-square test at 5%. Of 60 trained students, 48 pass and 12 fail; of 40 untrained students, 22 pass and 18 fail. Show expected frequencies and the statistic. Use $\chi^2_{0.05,1} = 3.841$; do not apply a continuity correction. (5)
- 12) Three independent normal populations have equal variances. Samples are A: 4, 5, 6, B: 7, 8, 9, C: 10, 11, 12. The between-group sum of squares is 54 and within-group sum of squares is 6. Complete the one-way ANOVA degrees of freedom, mean squares and *F* statistic. Use $F_{0.05;(2,6)} = 5.143$ and state your conclusion. (3)
- 13) Distinguish a Type I error from a Type II error in the context of the filling-machine test in Question 8. (2)

End of question paper